Grade 12 Foundations for College Math
Final Exam Key

First Name: _____**KEY**_____

Last Name: _____

Directions:

- Show all work for full marks. Round dollar amounts to the nearest cent.
- The exam is designed to be completed in 75 minutes.

For grading use only

Page:	2	3	4	5	6	7	Total
Points:	7	7	18	20	30	18	100
Score:							

Part A — Multiple Choice (20 marks)

1. (1 point) The future value of an annuity formula $FV = PMT \cdot \dfrac{(1+i)^n - 1}{i}$ is used when:
   - A. Finding the present lump sum needed now
   - **B. Finding the accumulated value of regular deposits**
   - C. Calculating simple interest
   - D. Finding the monthly payment on a loan

2. (1 point) \$500 is deposited monthly into an account at 6% per year compounded monthly for 10 years. The number of compounding periods is:
   - A. 10
   - B. 60
   - **C.** 120
   - D. 72

3. (1 point) Which formula gives the monthly mortgage payment?
   - A. $FV = PMT \cdot \dfrac{(1+i)^n - 1}{i}$
   - **B.** $PV = PMT \cdot \dfrac{1-(1+i)^{-n}}{i}$**, solved for** PMT
   - C. $A = P(1+i)^n$
   - D. $I = Prt$

4. (1 point) A \$200 000 mortgage at 5% per year compounded semi-annually, 25 years, monthly payments — the effective monthly rate is:
   - A. $\dfrac{0.05}{12}$
   - **B.** $(1+0.025)^{1/6} - 1$
   - C. $\dfrac{0.05}{2}$
   - D. $0.05 \div 25$

5. (1 point) A data set is best modelled by a quadratic function when:
   - A. The data increases by a constant amount each period
   - **B. The data increases then decreases (or vice versa), forming a curve**
   - C. The data doubles each period
   - D. The data oscillates regularly

6. (1 point) The correlation coefficient $r = -0.88$ indicates:
   - A. Weak positive correlation
   - B. No correlation
   - **C. Strong negative correlation**
   - D. Weak negative correlation

7. (1 point) For the exponential model $y = 3(2)^x$, the doubling time is:
   - A. 2

B. 1

C. 3

D. 0.5

8. (1 point) The sinusoidal model $y = 4\sin(2x) + 3$ has amplitude:

A. 2

B. 4

C. 3

D. 7

9. (1 point) The Cosine Law is used when you know:

A. Two angles and a side

B. One side and all angles

C. Two sides and the included angle (SAS) or all three sides (SSS)

D. Two sides and a non-included angle

10. (1 point) A cylinder with $r = 2$ m and $h = 5$ m has total surface area:

A. 14π m^2

B. 20π m^2

C. 28π **m**^{2}

D. 40π m^2

11. (1 point) Two similar cones have linear scale factor $1:3$. Their volume ratio is:

A. $1:3$

B. $1:6$

C. $1:27$

D. $1:9$

12. (1 point) A ship sails on bearing S70°W for 40 km. Its southward displacement is:

A. $40\cos 70^\circ$

B. $40\cos 70^\circ$ **south (approximately** 13.7 km**)**

C. $40\sin 70^\circ$ south

D. 40 km south

13. (1 point) For a normal distribution with $\mu = 80$ and $\sigma = 10$, the z-score for $x = 95$ is:

A. 0.5

B. 1.0

C. 1.5

D. 2.0

14. (1 point) About 95% of data in a normal distribution lies within:

A. $\mu \pm \sigma$

B. $\mu \pm 2\sigma$

C. $\mu \pm 3\sigma$

D. $\mu \pm 0.5\sigma$

15. (1 point) A survey of 900 people has a margin of error of $\pm 3.3\%$ at 95% confidence. To get a margin of error of $\pm 1.65\%$, the sample size needed is:
    A. 1 800
    B. 2 700
    C. 3 600
    D. 450

16. (1 point) The magnitude of vector $\vec{v} = (-5, 12)$ is:
    A. 7
    B. 10
    C. 13
    D. 17

17. (1 point) The perpendicular bisector of segment AB is the locus of points:
    A. On line AB
    B. Equidistant from A and B
    C. Equidistant from A and a fixed line
    D. Within a fixed distance from A

18. (1 point) An inscribed angle of $55°$ subtends an arc. The central angle subtending the same arc is:
    A. $27.5°$
    B. $110°$
    C. $55°$
    D. $125°$

19. (1 point) To prove $PQRS$ is a parallelogram using midpoints of diagonals, you verify:
    A. All sides are equal
    B. The diagonals are perpendicular
    C. The midpoints of both diagonals are the same point
    D. All angles are $90°$

20. (1 point) The shoelace formula gives the:
    A. Perimeter of a polygon
    B. Area of a polygon given its vertices
    C. Length of the diagonals
    D. Number of sides

Part B — Short Answer (40 marks)

21. (8 points) **Financial Mathematics.**
    (a) (4 points) Mei contributes \$400 per month to an RRSP at 6% per year compounded monthly for 30 years. Find the total value at retirement and the total amount she contributed. How much is interest?

$$(FV = PMT \cdot \frac{(1+i)^n - 1}{i})$$

(b) (4 points) Mei then uses the RRSP to fund a monthly pension. She needs \$2 500 per month for 20 years at 4% per year compounded monthly. Find the present value required at retirement. Does her RRSP cover it?

$$(PV = PMT \cdot \tfrac{1-(1+i)^{-n}}{i})$$

Solution: (a) $i = \dfrac{0.06}{12} = 0.005$, $n = 30 \times 12 = 360$

$$FV = 400 \cdot \frac{(1.005)^{360} - 1}{0.005} = 400 \cdot \frac{6.0226 - 1}{0.005} = 400 \cdot \frac{5.0226}{0.005} = 400 \times 1004.52 \approx \$\mathbf{401\,808}$$

Total contributed: $400 \times 360 = \$144\,000$

Interest earned: $\$401\,808 - \$144\,000 = \$\mathbf{257\,808}$

(b) $i = \dfrac{0.04}{12} \approx 0.003\overline{3}$, $n = 20 \times 12 = 240$

$$PV = 2500 \cdot \frac{1 - (1.003\overline{3})^{-240}}{0.003\overline{3}} = 2500 \cdot \frac{1 - 0.4493}{0.003\overline{3}} = 2500 \cdot \frac{0.5507}{0.003\overline{3}} \approx 2500 \times 165.02 \approx \$\mathbf{412\,550}$$

Her RRSP of \$401,808 does not quite cover the required \$412,550 — she is short by about \$10,742.

22. (8 points) **Modelling with Functions.** The monthly average temperature (°C) for a city is given below:

Month	J	F	M	A	M	J	J	A	S	O	N	D
Temp (°C)	−8	−6	0	8	15	20	23	22	16	9	1	−5

(a) (2 points) Identify the maximum and minimum temperatures. Find the amplitude and vertical shift of a sinusoidal model.

(b) (3 points) The period is 12 months. Find k in $y = a\sin(k(x-d)) + c$. Write a complete sinusoidal equation for the temperature, using a sine function with phase shift $d = 4$ (April as the starting rise). Use $x = 1$ for January.

(c) (3 points) Use your model to predict the temperature in month 8 (August) and month 3 (March). Compare to the actual values and comment.

Solution: (a) Max = 23°C (July), Min = −8°C (January).

Amplitude: $a = \dfrac{23 - (-8)}{2} = \dfrac{31}{2} = \mathbf{15.5°C}$

Vertical shift: $c = \dfrac{23 + (-8)}{2} = \dfrac{15}{2} = \mathbf{7.5°C}$

(b) $k = \dfrac{2\pi}{12} = \dfrac{\pi}{6}$

Model (using sine with $d = 4$, since the temperature is near the midline rising around April):

$$y = 15.5\sin\left(\frac{\pi}{6}(x-4)\right) + 7.5$$

(c) August: $x = 8$: $y = 15.5\sin\left(\dfrac{\pi}{6}(4)\right) + 7.5 = 15.5\sin\left(\dfrac{2\pi}{3}\right) + 7.5 = 15.5(0.866) + 7.5 \approx \mathbf{20.9°C}$

Actual: 22°C. Difference: 1.1°C — close.

March: $x = 3$: $y = 15.5\sin\left(\frac{\pi}{6}(-1)\right) + 7.5 = 15.5(-0.5) + 7.5 = -7.75 + 7.5 \approx \mathbf{-0.25°C}$

Actual: 0°C. Difference: 0.25°C — very close.

The sinusoidal model fits the data well. Slight discrepancies occur because real temperature patterns are not perfectly sinusoidal.

23. (8 points) **Trigonometry & Geometry.**

(a) (4 points) From the roof of a building (height h), a surveyor measures the angle of elevation to the top of a taller building as 22° and the angle of depression to the base of that building as 35°. The horizontal distance between the buildings is 80 m. Find h (the height of the first building) and H (the total height of the second building).

(b) (4 points) A storage tank is a cylinder with a cone on top. The cylinder has radius 4 m and height 6 m. The cone has the same base radius and slant height 5 m. Find the total surface area and total volume.

Solution: (a) Let $d = 80$ m (horizontal distance), h = height of first building.

From angle of depression to the base: $\tan 35° = \frac{h}{80}$

$$h = 80\tan 35° = 80(0.7002) \approx 56.0\,\text{m}$$

From angle of elevation to the top of second building: Let x = height of second building above the surveyor's level. $\tan 22° = \frac{x}{80} \Rightarrow x = 80\tan 22° = 80(0.4040) \approx 32.3\,\text{m}$

Total height of second building: $H = h + x = 56.0 + 32.3 \approx 88.3\,\text{m}$

(b) Cone: $r = 4$ m, slant height $l = 5$ m, so cone height $= \sqrt{5^2 - 4^2} = 3$ m.

Total Surface Area:

- **Circular base of cylinder:** $\pi(4)^2 = 16\pi$
- **Lateral surface of cylinder:** $2\pi(4)(6) = 48\pi$
- **Lateral surface of cone:** $\pi r l = \pi(4)(5) = 20\pi$
- **(No circular top on cylinder — the cone sits there)**

$SA = 16\pi + 48\pi + 20\pi = 84\pi \approx \mathbf{263.9\ m^2}$

Total Volume: $V_{\text{cyl}} = \pi(4)^2(6) = 96\pi$; $V_{\text{cone}} = \frac{1}{3}\pi(4)^2(3) = 16\pi$

$V = 96\pi + 16\pi = 112\pi \approx \mathbf{351.9\ m^3}$

24. (8 points) **Statistics.** The lifespans (years) of a species of parrot are normally distributed with $\mu = 35$ and $\sigma = 6$.

(a) (3 points) Find the percentage of parrots that live: (i) between 29 and 41 years; (ii) more than 47 years; (iii) less than 23 years.

(b) (3 points) Find the lifespan at the 75th percentile ($z_{0.75} = 0.674$) and the lifespan below which 10% of parrots fall ($z_{0.10} = -1.282$).

(c) (2 points) A wildlife researcher surveys 100 parrots and finds a mean lifespan of 36.2 years. The margin of error at 95% confidence is $\pm\dfrac{1.96 \times 6}{\sqrt{100}}$. Find the confidence interval and interpret it.

Solution: (a) (i) $29 = 35 - 6 = \mu - \sigma$ **and** $41 = 35 + 6 = \mu + \sigma$**. By the 68-95-99.7 rule: 68%**

(ii) $47 = 35 + 2(6) = \mu + 2\sigma$**. Above** $\mu + 2\sigma$**:** $\dfrac{100-95}{2} = \mathbf{2.5\%}$

(iii) $23 = 35 - 2(6) = \mu - 2\sigma$**. Below** $\mu - 2\sigma$**: 2.5%**

(b) 75th percentile: $x = \mu + z\sigma = 35 + 0.674(6) = 35 + 4.04 =$ **39.0 years**

10th percentile: $x = 35 + (-1.282)(6) = 35 - 7.69 =$ **27.3 years**

(c) $ME = \dfrac{1.96 \times 6}{\sqrt{100}} = \dfrac{11.76}{10} = 1.176$

Confidence interval: $36.2 \pm 1.176 \Rightarrow (\mathbf{35.02},\ \mathbf{37.38})$ **years**

Interpretation: We are 95% confident that the true mean lifespan of all parrots of this species is between 35.0 and 37.4 years.

25. (8 points) **Applications of Geometry.**

(a) (4 points) A ship leaves port O and sails on bearing N50°E for 120 km to point A. It then sails on bearing S30°E for 90 km to point B. Find the distance OB and the bearing from O to B.

(b) (4 points) The vertices of quadrilateral $PQRS$ are $P(0,1)$, $Q(4,3)$, $R(6,0)$, $S(2,-2)$. Use midpoints of diagonals to prove $PQRS$ is a parallelogram. Then find its area using the shoelace formula.

Solution: (a) The angle at A **between the two legs:**

Bearing N50°E (first leg) and S30°E (second leg):

The first leg arrives at A **from the direction S50°W. The second leg departs on S30°E. The angle between these two directions at** A**:** $\angle OAB = 180° - 50° - 30° = 100°$

By Cosine Law:

$$OB^2 = 120^2 + 90^2 - 2(120)(90)\cos 100°$$

$$= 14\,400 + 8\,100 - 21\,600(-0.1736) = 22\,500 + 3\,750 = 26\,250$$

$$OB = \sqrt{26\,250} \approx 162.0\,\text{km}$$

For bearing: $\dfrac{\sin(\angle AOB)}{90} = \dfrac{\sin 100°}{162.0}$

$\sin(\angle AOB) = \dfrac{90 \times 0.9848}{162.0} = 0.5463 \Rightarrow \angle AOB \approx 33.1°$

Bearing from O **to** B**: N50°E rotated** $33.1°$ **clockwise = N83°E (approximately).**

(b) Midpoint of PR**:** $\left(\dfrac{0+6}{2}, \dfrac{1+0}{2}\right) = (3, 0.5)$

Midpoint of QS**:** $\left(\dfrac{4+2}{2}, \dfrac{3+(-2)}{2}\right) = (3, 0.5)$ ✓

Diagonals bisect each other, so $PQRS$ **is a parallelogram.**

Shoelace formula:

$$\textbf{Area} = \tfrac{1}{2}|x_P(y_Q - y_S) + x_Q(y_R - y_P) + x_R(y_S - y_Q) + x_S(y_P - y_R)|$$

$$= \frac{1}{2}|0(3-(-2)) + 4(0-1) + 6((-2)-3) + 2(1-0)|$$

$$= \frac{1}{2}|0 + (-4) + (-30) + 2| = \frac{1}{2}|-32| = \mathbf{16\ square\ units}$$